

COMBINATORICS

EXAM 1

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1. Answer the following using simple counting reasoning

- (a) How many different 7-digit numbers (non-negative integers) are there? (Leading zeros are not allowed).
- (b) How many even 7-digit numbers are there?
- (c) How many 7-digit numbers are there with exactly one 7?
- (d) How many 7-digit numbers read the same when you read them forward or backward, i.e. are palindromic, e.g. 6548456 is one such integer.
- (e) How many 7-digit numbers are there that are divisible by 4?
- (f) How many times is the digit 7 written when listing the numbers between 1 and 1,000,000?

Solution. (a) If we consider the number simply as a string of characters, and we allowed “0” at the beginning, then we would have 10^7 solutions, but for our leading digit we omit the availability of 0. So our count becomes 9×10^6 .

(b) To get the number of 7-digit even numbers with the same rules, we simply take the result from (a) and divide by 2. So we get our count to be $(9 \times 10^6)/2 = 4.5 \times 10^6$.

(c) Let’s approach the problem in cases. Case 1 (the 7 is the first digit): the other 6 digits are anything but 7 which is 9^6 options. Case 2 (the 7 is in one of the other 6 positions): we start with the first digit which is 8 choices, because it can’t be 0 or 7. Next we fill the remaining 5 non-7 digits in 9^5 ways. Lastly, we have to choose which of the 6 digits is our 7, and there are 6 ways to do that. So the count of 7 in one of the other 6 positions comes out as $8 \cdot 9^5 \cdot 6$.

(d) Considering the palindromes, we need only consider half the word because the other half mirrors it, excepting the middle character. Thus, our count for ways to have a palindrome becomes 9×10^3 .

(e) At first glance I went about simply dividing (a) by 4, but that over simplifies the task at hand and omits some subtleties. We begin by noticing that if a number is divisible by 4 then the last two digits form a number that is divisible by 4.

So let’s break the number down into $d_1d_2d_3d_4d_5d_6d_7$. To build our number we chose the first digit d_1 between 1 and 9 to remain a 7-digit number. The middle digits can be

anything from 0 to 9 which gives us 10^4 choices. Lastly, the last two digits must be in the set $\{00, 04, 08, \dots, 96\}$ which has 25 elements. So the number of choices we have to build our number is $9 \cdot 25 \cdot 10^4 = 2250000 = 2.25 \times 10^6$

(f) Notice that 1,000,000 can not contain a 7 so we only concern ourselves with numbers of length 6. In each of the 6 positions a 7 occurs in 10^5 numbers, 6×10^5 . And, because 0 and 1,000,000 don't contain the digit 7 we're good with our count being 6×10^5 . $\square$

2. How many 7-letter words can be constructed by using the 26 letters of the alphabet if each word contains 3, or 4 vowels? It is understood that there is no restriction on the number of times a letter can be used in a word.

Solution. Let $\{a, e, i, o, u, y\}$ be our set of vowels. The number of 7-letter words with 3 vowels is

$$\binom{7}{3} \cdot 6^3 \cdot 20^4.$$

The number of 7-letter words with 4 vowels is

$$\binom{7}{4} \cdot 6^4 \cdot 20^3.$$

So our count follows as:

$$\# = \binom{7}{3} \cdot 6^3 \cdot 20^4 + \binom{7}{4} \cdot 6^4 \cdot 20^3 = 1572480000$$

$\square$

3. Let n and k be positive integers. Give a combinatorial proof that

$$\sum_{k=1}^n k \binom{n}{k}^2 = n \binom{2n-1}{n-1}$$

Proof. We can reduce this problem to a question about a committee with men and women. And we want to choose n committee members from a pool of n men and n women, and we want a woman chairperson. The right hand side is clearly a count of just this. For the left hand side consider a committee we choose k women from n choices, $\binom{n}{k}$. Then, we count the $n-k$ men we chose from the n men $\binom{n}{n-k} = \binom{n}{k}$. All we are left to do is select woman who is supposed to chair the committee, and there are n ways to accomplish this task. And thus we clearly have that:

$$\sum_{k=1}^n k \binom{n}{k}^2 = n \binom{2n-1}{n-1}$$

$\square$

4. In how many ways can we select a set of m marbles from a collection of $2m$ marbles, where m of them are identical and the remaining m are all distinct?

Solution. Let the collection of marbles look like $\{\underbrace{l, l, \dots, l}_{m\text{-terms}}, m_1, m_2, \dots, m_m\}$. Choose r distinct marbles where $0 \leq r \leq m$ in $\binom{m}{r}$ ways. Then we must choose the identical marbles in $m - r$ ways, for which there is only 1 way to do this since they are all identical. Thus, the number of ways to do our selection generally is:

$$\sum_{r=0}^m \binom{m}{r} = 2^m.$$

□

5. Show that if $n + 1$ distinct integers are chosen from the set $\{1, 2, \dots, 3n\}$, then there are always two which differ by at most 2.

Proof.

□

6. Use the pigeonhole principle to show that:

- (a) Any set of 7 distinct integers includes two integers a and b such that either $a + b$ or $a - b$ is a multiple of 10.
- (b) Any group of 20 students either has a subgroup of 4 mutual acquaintances, or a subgroup of 4 mutual strangers.

Solution.

□

7. What is the number of integral solutions of the equation

$$x_1 + x_2 + x_3 + x_4 = 30,$$

such that $x_1 \leq 1$, $x_2 \leq 2$, $x_3 \leq 3$, and $x_4 \leq 4$?

Solution.

□

8. Find the number of integer solutions to the equation

$$x_1 + x_2 + x_3 + x_4 = 20,$$

for which $1 \leq x_1 \leq 6$, $0 \leq x_2 \leq 7$, $4 \leq x_3 \leq 8$, and $2 \leq x_4 \leq 6$?

Solution.

□

9. Solve the recurrence relation

$$a_n = 5a_{n-1} + 2a_{n-2} - 24a_{n-3}, \quad n \geq 3$$

given that $a_0 = 1$, $a_1 = 2$, and $a_2 = 6$.

Solution.

□

10. Solve the recurrence relation $a_k = a_{k-1} + k + 6$, with $k \geq 1$ given that $a_0 = 5$.

Solution. □

11. Use generating functions to count the number of ways to collect \$15 from 20 different students, if each of the first 19 students can give either \$1 or nothing, and the twentieth student can give nothing, \$1, or \$5.

Solution. □

12. Consider an n -sided polygon (with n vertices and n sides).

- (a) Assuming no three diagonals intersect at a single point, count the points of intersection formed by the diagonals of this n -sided polygon.
- (b) How many line segments are formed by the intersection of points in part (a)?
- (c) One of the simplest and most elegant formulae in mathematics is Euler's polyhedron formula. A version for planar graphs states that if a planar graph with v vertices and e edges is drawn in the plane without any edge crossings and has r regions, then

$$r = e - v + 2$$

Use this formula together with your answers in parts (a) and (b) to count the number of regions formed by the diagonals of the n -sided polygon.

Solution. □

13. Show that any set of 3 distinct integers includes two integers x and y such that

$$F(x, y) = x^3y - xy^3$$

is a multiple of 10.

Proof. □

14. Use combinatorial reasoning to show that:

$$\binom{n}{r-1} = \sum_{i=0}^k (-1)^i \binom{k}{i} \binom{n+k+1}{r+k+1}$$

Proof. □

15. Find and verify a general formula, involving the Sterling Numbers of the second kind, for $1^n + 2^n + 3^n + \cdots + m^n$.

Solution. □